Bibliography

Dirac, P. A. M., *The Principles of Quantum Mechanics*, 4th Edition, (Oxford University Press, London, 1958).

Feynman, R. P. and A. P. Hibbs, *Quantum Mechanics and Path Integrals*, (McGraw-Hill, Inc. 1965).

Gürsey, Feza, Editor; *Group Theoretical Concepts and Methods in Elementary Particle Physics*, (Gordon and Breach, Science Publishers, Inc., New York, 1964).

Jordan, Thomas F., *Linear Operators for Quantum Mechanics*, 2nd Edition, (John Wiley & Sons, 1982).

Messiah, Albert, *Quantum Mechanics*, Vol. I & Vol. II, (North-Holland Publishing Co., Amsterdam; Interscience Publishers Inc., New York, 1962).

Riesz, F. and B. Sz-Nagy, *Functional Analysis*, translated by L. F. Boron, (F. Ungar Publishing Co., New York, 1955).

Tung, Wu-Ki, *Group Theory in Physics*, (World Scientific Publishing Co Pte Ltd., Singapore, 1985).

von Neumann, John, *Mathematical Foundations of Quantum Mechanics*, (Princeton University Press, Princeton, New Jersey, 1955).

Wigner, Eugene P., *Group Theory*, (Academic Press, New York, 1959).

Wybourne, Brian G., *Classical Groups for Physicists*, (John Wiley & Sons, 1974).

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